Proposal for Rolsa Technologies

The client, Rolsa Technologies, is a local well-established green technology company that is interested in the digitalisation of their offerings, to both help engage new customers and to help keep existing customers satisfied. Currently Rolsa Technologies offers: Solar panel installation and maintenance, Electric vehicle (EV) charging stations and Smart home energy management systems.

The client is looking for a digital solution that would provide customer with information about green energy products currently on the market and how to reduce their carbon footprint; allow customers to schedule consultations and installations and calculate their carbon footprint; have account registration to allow customers to manage their consultations and data; have accessibility features to support a wide range of users; and have a tool for calculating and tracking energy usage.

I am proposing to develop an interactive website that would include all of the above features that the client, Rolsa Technologies, and some more. Below I will be explaining and justifying my proposal.

Business Context:

Business Considerations:

Aspect	Details
Client Needs	**Provide customer with information about green energy products currently on the market** – To meet this need the website will need to showcase a variety of green energy products that are currently on the market. The website should highlight Rolsa's own products for the markets it specialises in: Solar panel installation and maintenance, Electric vehicle charging stations and smart home energy management systems. But it should also highlight other product also on the market that I have identified in research including: Heat pumps and Domestic wind turbines. On the website, I will have a dedicated products page to provide all potential customers with information about both Rolsa Technologies and others green energy products that are currently on the market.
	Provide customer with information about how to reduce their carbon footprint – To meet this need the website will need to provide a variety of ways that people are able to reduce their own carbon footprint. The website should highlight ways recommended by the United Nations' Actions for a Healthy Planet that including: Save energy at home, change your home's source of energy, walk bike or take public transport etc. I also believe to meet this need the website should offer users a quiz to score how "eco-friendly" their actions currently are and give them personalised feedback for how to lower their carbon footprint. To meet this need I will create two dedicated pages one for the static information and one for the quiz.
	Allow customers to schedule consultation and installation – To meet this need the website should allow staff users to set available consultations and installations and allow customer users to select one of the available options.
	Allow customers to calculate their carbon footprint – To meet this need the website should allow all users to use a tool to calculate their own personal carbon footprint. Through research I have found a Carbon Interface API which I could use to save time to focus on other features while still ensuring a high quality and high accuracy experience for all users. Carbon Interface | The Carbon Estimation Platform https://www.carboninterface.com/
	Have account registration to allow customers to manage their consultations and data – To meet this need the website should allow all users to both register and sign into the website with an account, this will allow customer users to schedule both consultations and installations. To fully meet this need, I will also include staff users who will be able to add available booking slots for customers to book both consultations and installations and they will also be able to view all bookings. Additionally, to fully meet this need, I will also include an admin user who will be able to access all the databases information.
	Have accessibility features to support a wide range of users – To meet this need the website should allow users to change accessibility features

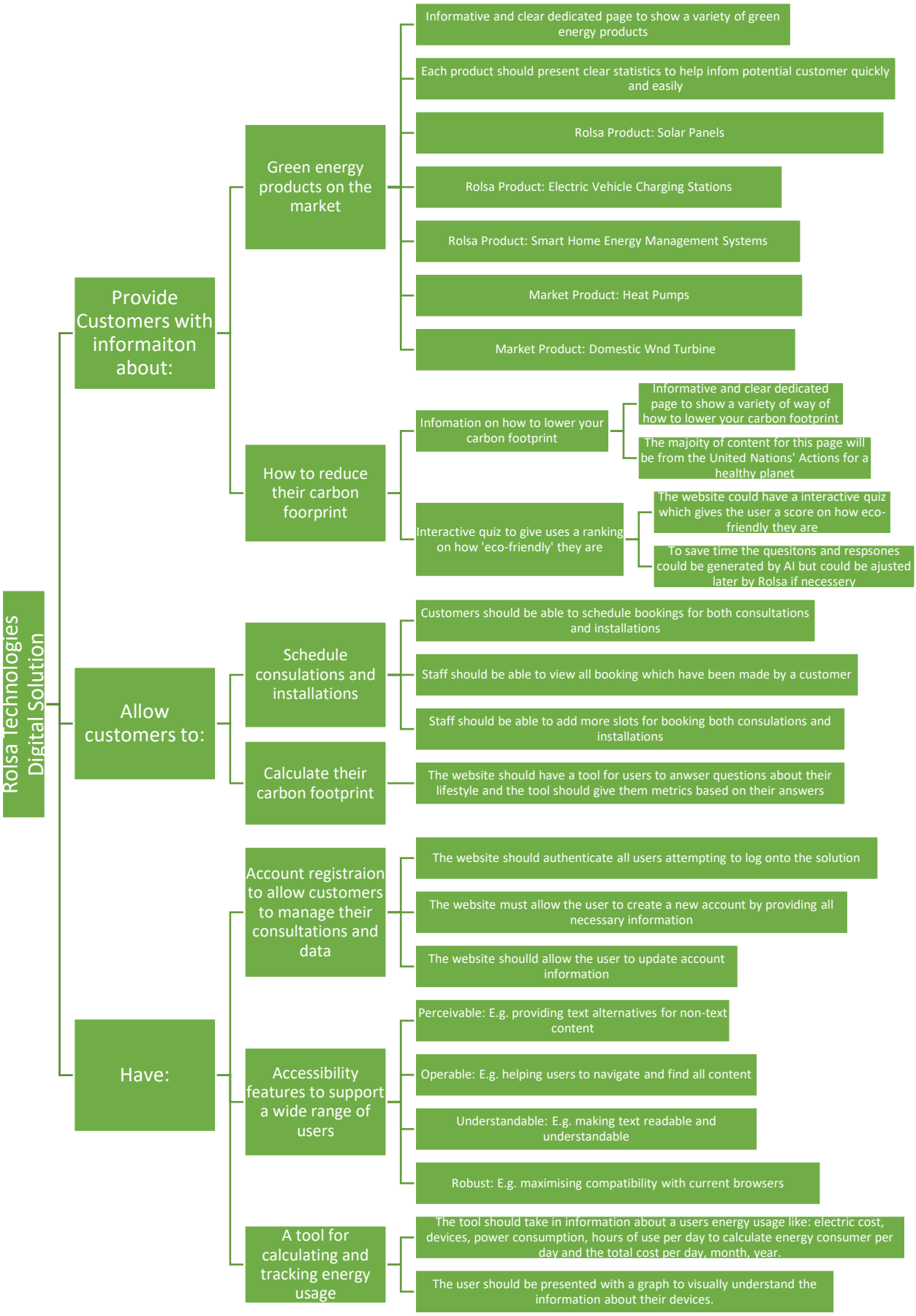
	to help meet W3C's WCAG guidelines by ensuring that all content on the website is perceivable, operable, understandable and robust.
	Have a tool for calculation and tracking energy usage – To meet this need the website should allow the user to enter information about their devices and the energy consumption to calculate and track their energy usage. This will have its own dedicate page and inputs will be saved to a user's account so that they can track differences over time.
Industry Relevance	There is a growing demand for more eco-friendly technologies and more people need to be aware of the positive impact these technologies are having on the planet. Offering free tools like calculating their carbon footprint and for calculating and tracking their energy usage it will make Rolsa Technologies stand out from their competition. It will also bring social benefits to Rolsa due to helping to improve their public image by offering these tools for free rather than a charge.
Current Challenges	The following are challenges that are currently facing the green technology industry: • Economic – High initial investment costs for green technology could be a barrier from new customers • Lack of Widespread Adoption – Challenges encouraging users to adopt the technology and integrating it into existing systems and infrastructures • Regulatory Challenges – Regulations can vary by region can could impede the development and implementation of green technologies.
Target Audience	The following are key user groups and will be explained in more detail in the empathy map/user stories section below: Customers: Existing customers of Rolsa. Customers: New customers to Rolsa with eco-friendly mindsets Staff: Use the new solution to manage and view all bookings
Value Proposition	This proposal will benefit all key user groups. It will benefit both new and existing customers by allowing them to view all products that Rolsa currently offer, give them tools for calculating their impact on the planet and allow them to book consultations and installations at a time that suits them. It will benefit staff at Rolsa by allow they to easily view and add slots for bookings of consultations and installations. It will benefit Rolsa as a business due to streamlining their own internal processes.

Empathy map/User Stories:

As a [user]	User Need (I want…)	Action (I should be able to…)	Acceptance Criteria
Existing Customer of Rolsa	I want to be able to view what other product Rolsa offer and see the benefits it could bring to me. I want to be able to calculate and track my energy usage to see how I can lower my energy usage.	I should be able to easily and quickly find other products that Rolsa have, and also be able to view what other products they recommend. I should be able to easily and quickly calculate and track my energy usage and be able to view a graph showing me the information.	• Informative page of information highlighting what products Rolsa and others offer and see the benefits they will bring. • A tool to calculate a user's energy usage over time to see what devices are using the most power and have the highest cost. • The above suggested above should have a graph to show these metrics visually to the user.
New Customers which have found Rolsa	I want to be able to view all products that Rolsa have and see other ones currently on the market that they recommend. I want to be able to view information about how to lower my carbon footprint. I want to be able to calculate my carbon footprint.	I should be able to easy and quickly find other products that Rolsa have, and also be able to view what other products they recommend. I should be able to easily and quickly find information about how to lower my carbon footprint. I should be able to use a tool to easily and accurately calculate my carbon footprint.	• Informative page of information highlighting what products Rolsa and others offer and see the benefits they will bring. • Informative page of information highlighting ways people can lower their own carbon footprint. • A tool to calculate a user's own personal carbon footprint. • The above tool should be accurate to ensure we are not misinforming the user.
Rolsa Staff	I want to be able to view all bookings for consultations and installations which have been made.	I should be able to easily and quickly find all bookings of consultations and installations.	• A dedicated staff page allowing staff to view of booked consultations and installations. • A dedicated staff page to allow staff to add

	I want to be able to add available slots for booking consultations and installations for customer to book.	I should be able to easily and quickly add more slots for booking consultations and installations.	more slots for booking consultations and installations.

Decomposition of the Problem:



Functional Requirements:

Requirement	Justification	How	Priority
The website must authenticate all users attempting to log onto the solution.	To meet the client's need of 'account registration' while meeting relevant privacy regulation and best-practices.	The solution will ensure passwords and emails match those stored in the user database.	Must
The website must allow the user to create a new account by providing all necessary information.	To meet the client's need of 'account registration to allow customers to manage their consultations and data'.	The solution will allow all users to both register an account and save this information to the user database. The password should be hashed.	Must
The website should allow the user to update account information.	To meet the client's need of 'account registration' while meeting relevant privacy regulation and best-practices.	The solution should allow all users to edit any information they entered in the registration process, and then change the data stored in the database.	Should
The website should clearly present information about Rolsa's green energy products to potential customers	To meet the client's need of 'provide customers with information about green energy products currently on the market'.	The solution will present information, details and benefits of using the green energy product.	Must
The website should clearly present information about other green energy products currently on the market	To meet the client's need of 'provide customers with information about green energy products currently on the market'.	The solution will present information, details and benefits of using the green energy product.	Must
The website should clearly present information about how people can lower their carbon footprint	To meet the client's need of 'provide customers with information about how to reduce their carbon footprint'.	The solution will present information, details and advice on how to lower your carbon footprint.	Must
The website could feature an interactive quiz to give users a score on how 'eco-friendly' there are and give them personalised tips on how to improve	To meet the client's need of 'provide customers with information about how to reduce their carbon footprint'.	The solution could present an interactive quiz to users, using a form, based on their answers they can be given feedback based on their worst answers.	Could
The website will allow customer users to schedule bookings for both consultations and installations.	To meet the client's need of 'allow customers to schedule consultations and installations'.	The solution will allow customer users to use a form and book their appointment. The form will get information regarding booking slots from the database.	Will

The website will allow customers to view their bookings of both consultations and installations	To meet the client's need of 'allow customers to manage their consultations'.	The solution will allow customer users to view their upcoming and previous appointments. The website will get this information from the bookings database.	Will
The website will allow customers to manage/cancel their bookings of both consultations and installations	To meet the client's need of 'allow customers to manage their consultations'.	The solution will allow customer user to manage and cancel their bookings. The website will update the booking database if any information change.	Will
The website will allow staff users to easily and quickly view all booking of both consultations and installations in a logical order	To meet the client's need of 'allow customers to schedule consultations and installations'.	The solution will allow staff users to view all bookings in the staff dashboard. The website will get this information from the database.	Will
The website will allow staff users to easily and quickly add slots for bookings of both consultations and installations	To meet the client's need of 'allow customers to schedule consultations and installations'.	The solution will allow staff users to add bookings slots for customer to book appointments. The website will add slots to the booking database.	Will
The website will allow users to calculate their energy usage based on questions they have answered	To meet the client's need of 'a tool for calculating and tracking energy usage'.	The solution will have a dedicated page allowing users to enter information about their energy usage. The information will be saved to a database and could be present to the user as a graph.	Will

Non-Functional Requirements:

Requirement	Justification	How	Priority
The website must provide a page response time of 2 seconds or less in a Chrome desktop browser	To ensure the digital solution is of a high standard and users aren't put off by slow load times.	If load time are becoming an issue the following could be done to improve load time: • Compress images, • Avoid using many animations, • Reduce the number of redirects	Must
The system must be secure from unauthorised access, for example mitigating against SQL injection	To ensure the digital solution is of a high standard and user's personal data is not lost through malicious attacks.	The solution will use input validation to ensure not SQL code can be entered through an input within any form.	Must
The website must be able to scale up and scale down as Rolsa's demand changes. For example, adding more products into the products page	To ensure the digital solution is of a high standard and it meets Rolsa's changing and growing business needs.	The solution will be easy to update by following coding best practices to ensure future content can be uploaded. For example, my Python code will follow PEP8 Guidelines (Python Enhancement Proposal 8)	Should
The website should be available for all users when needed with very little down time	To ensure the digital solution is of a high standard and users aren't disappoint with the performance of the website.	When the solution is hosted Rolsa will want to ensure they use a reliable hosting company to reduce down time of the website.	Should
The website must be easy to maintain and update	To ensure the digital solution is of a high standard and future updates are easy to implement onto the digital solution.	The solution will be easy to update by following coding best practices like PEP8. To ensure good maintainability I will ensure good readability, modularity and documentation.	Must
The website should be compatible with all major browsers and devices	To ensure the digital solution is of a high standard and users aren't disappoint with the performance of the website on a variety of devices and browsers.	When the solution is hosted more browser and device testing can be completed to ensure fully functionality across entire digital solution for Rolsa.	Should
The website should be perceivable to all users for example by providing text alternatives for non-text content	To meet the client's need of 'accessibility features to support a wide range of users' and meeting W3C's WCAG guideline of 'perceivable'.	The solution will be perceivable, by: • Providing text alternatives for non-text content • Making it easier for users to see and hear content	Must

The website should be operable to all users for example by helping users to navigate and find all content	To meet the client's need of 'accessibility features to support a wide range of users' and meeting W3C's WCAG guideline of 'operable'.	The solution will be operable, by: • Making all functionality available from a keyboard • Giving users enough time to read and use content • Helping users navigate and find content	Must
The website should be understandable to all user for example by making text readable and understandable	To meet the client's need of 'accessibility features to support a wide range of users' and meeting W3C's WCAG guideline of 'understandable'.	The solution will be understandable, by: • Making text readable and understandable • Making content appear and operate in predictable ways • Helping users to avoid and correct mistakes	Must
The website should be robust to all users for example by being able to handle user errors	To meet the client's need of 'accessibility features to support a wide range of users' and meeting W3C's WCAG guideline of 'robust'.	The solution will be robust, by: • Maximising compatibility with current and future user tools • Able to handle all user errors	Must

Key Performance Indictors (KPIs):

These metrics help to assess the website's effectiveness, usability and impact.

Type	KPI	Justification & how
Code Quality	Include clear comments for every Class and Function throughout all Python & JavaScript files. (Measured as %)	Ensuring frequent comment through the code will ensure good maintainability through readability. At the end of each week or end of project I can calculate the percentage of classes and function that do and do not have comments.
Testing Quality	Perform at least 20 tests on newly added features currently in development each week	Ensuring frequent tests are performed each day on features that are currently in development each week. Ensuring new features do not impact previous features by frequently completing end-to-end tests. At the end of each week a review can be done to see how many tests were completed in that week.
	Ensure at least 90% of failed tests carried out are resolved before the end of each day	Ensuring that any of the frequent tests that fail are fixed and corrected before the end of each day. This will ensure any found error are corrected as soon as possible and reduce the risk of effecting other features currently in development. At the end of each week a review can be done to see how many tests failed each done and when they were fixed, to calculate a percentage of test that were fixed on same day they are found.
	At least 5 tests done for each webpage for end-to-end testing	Throughout the project testing each page to allow any accidental changes, that were made at a later point, found as a bug to ensure the final product is fully functional meeting all of the above requirements, from end-to-end of the website's user experience. At the end of each week a review can be done to see how many end-to-end tests were completed to keep track of progress on this KPI
System Performance	Page load time must be no more than 2 seconds on average	Ensuring the user experience is of a high quality, this will help to ensure

		the user doesn't change to a competitor's website. Through the development process Lighthouse report can be completed to track each page's loading time to review how on track I am for this KPI.
	Query and database update time must be no more than 2 seconds on average	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website. Through the development process Lighthouse report can be completed to track each page's loading time to review how on track I am for this KPI.
	At least 95% of both functional and non-functional requirement must be met by the end of the development stage.	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website. Through the development process Lighthouse report can be completed to track each page's loading time to review how on track I am for this KPI.
Productivity	Spend 10 hours a week working on development, testing and documentation	Ensuring the project is able to be completed within the specified time frame to a high quality. Through the development process keeping track of how many hours are spent of the development, testing and documentation of the website to ensure this KPI is met.
Client/User Satisfaction	Client/User satisfaction ratings to stay above an average of 75%	During the evaluation and user feedback stage, asking test users for a satisfaction rating to they stay above a rating of 75% on average across all test uses. Following the development stage asking test users to complete a feedback questionnaire to gather this metric.
	Less than 10% of users report a problem with the solution	During the evaluation and user feedback stage, asking test users to report any problem they encounter. Following the development stage asking test users to complete a feedback questionnaire to gather this metric.

User Acceptance Criteria:

These are conditions the website must meet to be accepted by the stakeholders and users.

Type	Criteria	Justification & how
Functionality	Users can navigate the website easily via a clear navigation menu and links	Ensuring all users are able to easily fully navigate the website through clear navigation menu and links spread through the entire website. The website will have a traditional navigation bar at the top of the user's screen to ensure all users are able to navigate around the website. The navbar will follow web conventions so the user can know what to expect and how to functionally use the navigation bar.
	All forms submit successfully, validating the inputs and send confirmation to the user	Ensuring the digital solution frequently communicates with the user to ensure the user is able to understand why something is working or what has something done. The website will use an alert bar to have this frequent communication with the user, providing them with feedback if a form is incorrectly submitted or when the form is successfully submitted.
Performance	Website loads within 3 seconds on both desktop and mobile	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website. Through the development process Lighthouse report can be completed to track each page's loading time to review how on track I am for this user acceptance criteria.
	The site functions correctly on major browser including: Chrome, Firefox, Safari and edge	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website.

		When the solution is hosted more browser and device testing can be completed to ensure fully functionality across entire digital solution for Rolsa.
Accessibility	The website meets WCAG compliance for screen readers and keyboard navigation	Ensuring the website meets the user need of "accessibility features to support a wide range of users". As explained above the website will follow W3C's WCAG guidelines to ensure the website is fully perceivable, operable, understandable and robust.
	Colour contrast and font sizes are appropriate for readability	Ensuring the website meets the user need of "accessibility features to support a wide range of users". As explained above the website will follow W3C's WCAG guidelines to ensure the website is fully perceivable, operable, understandable and robust.
Mobile Responsiveness	The site adjusts properly on screens from 320px (mobile) to 1440+ px (desktop)	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website. When this solution is hosted, the website can be fully tested to ensure fully computable across a wide range of screen sizes.
	Navigation elements are touch-friendly and easily clickable	Ensuring the user experience is of a high quality, this will help to ensure the user doesn't change to a competitor's website. When this solution is hosted, the website can be fully tested to ensure fully computable across a wide range of screen sizes. During development mobile-friendly design practices can be completed to ensure the website is fully operable on mobile. For example, by ensuring all buttons are of adequate size to ensure they are touch-friendly.

Security & Data Protection	User data is encrypted and securely stored	Ensuring the website follow all regulatory and privacy regulations. To ensure the website is compliant with laws like the EU's GDPR and the UK's Data Protection Act. All sensitive data like passwords will be stored as hashes to ensure no one can view a user's password.
	Login and authentication systems are properly implemented	Ensuring the website follow all regulatory and privacy regulations. To ensure the website is compliant with laws like the EU's GDPR and the UK's Data Protection Act. A fully functioning authentication system will be implemented, to ensure the different levels of user access (no account, customer, staff and admin) can only access what is required for them to complete their role.
	The website follows GDPR best practices (e.g., clear cookie policy and opt-in consents0	Ensuring the website follow all regulatory and privacy regulations. To ensure the website is compliant with laws like the EU's GDPR and the UK's Data Protection Act. The website will follow the most common GDPR best practices for example by presenting to the user a clear cookie policy when they first view the website.
SEO & Discoverability	Meta titles, descriptions and structured data are properly set for search engines	To ensure the website have good SEO rating and discoverability to ensure the website can be fully successful for Rolsa. During development the Lighthouse report can be completed to calculate the website SEO score.
	Page URLs follow SEO friendly structure	To ensure the website have good SEO rating and discoverability to ensure the website can be fully successful for Rolsa. During development all URLs routes will be programmed using best practices and a logical way.

Description of the Proposed Solution:

My proposed solution to Rolsa Technologies is to create an interactive website, allowing a range of users with differing & varying accessibility requirements to:

- Be informed about green energy products both ones produced by Rolsa Technologies and others currently on the market through a dedicated “Our Solutions” page.
- Be informed about how to reduce their carbon footprint through a list of tips to lower a person’s environmental impact through a dedicated “Ways to Help” page.
- Be informed about how to reduce their carbon footprint through an interactive quiz, giving the user a score/rating of how environmentally friendly their day-to-day life is. The solution will also provide them with personalised feedback based on their least environmentally friendly choices. This will be presented to the user on a dedicated “Eco-Lifestyle Quiz” page.
- Book, schedule, view and manage their consultations and installations through an online account, through a dedicated “Bookings” and “Dashboard” pages.
- Calculate their carbon footprint, using a carbon footprint calculator API, through a dedicated “Carbon Calculator” page.
- Calculate and track their energy usage to identify what devices would be having the biggest environmental impact, through a dedicated “Energy Usage Tool” page.

Justification:

How my Solution meets the need of the client and users:

Stake Holder Group	What are their needs?	How will my solution fulfil their needs?
Customers	Be provided with information about green energy products currently on the market.	The website will have a dedicated "Products" page. This page will feature the following green energy products, highlight their benefits, and showcase their impacts through statistics for the following products: • Solar Panels, • Electric Vehicles (EV) Charging Stations, • Smart Home Energy Management Systems, • Heat Pumps, • Domestic Wind Turbine,
	Be provided with information about how to reduce their carbon footprint.	The website will have a dedicated "Ways to Help" pages. This page will highlight several ways people can help to lower their own carbon footprint and environmental impact. Some examples for tips that could feature on this page (based on the United Nations' Actions for a Healthy Planet list): • Save energy at home (linking to energy usage tool), • Walk, bike or take public transport, • Switch to an electric vehicle, • Reduce, reuse, repair and recycle, • Plant native species,
	Allow them to schedule, view and manage their consultations and installations.	The website will have several dedicated pages for this need, "Bookings" and "Dashboard". The "Bookings" page will contain a form allowing customer to book either a consultation or installation from a range of available slots. The "Dashboard" page will contain a list of all of the customer's bookings for both consultations or installations to view their upcoming bookings. This page will also allow the user to manage and cancel their appointment on the "Dashboard" page.
	Allow them to calculate their personal carbon footprint.	The website will have a dedicated "Carbon Calculator" page. This page will use an API (to ensure the project can be fully completed in the required amount of time) to calculate the users' personal carbon footprint.

		The API is fully free and accessible to a wide range of users, a necessity for the website.
	Use a tool to calculate and track their energy usage.	The website will have a dedicated “Energy Usage Tool” page. This page will contain a form which will allow the user to enter the following information, their home energy cost (1 kWh cost), then for each of their devices to enter name, power consumption (watts) and hours of use per day. From that information the website will calculate the energy consumer per day (kWh) to then calculate the cost per day. This information will then be presented to the user on a graph so that can visually understand what devices are consuming most of the electricity.
	Allow the to create an account which will manage their consultations and data.	The website will have a range of pages dedicated to the account system including “Register” and “Sign In”. These pages will allow the user to create an account and sign into their account respectively. Their passwords will be stored as hashes within the user database.
Staff	To view all scheduled consultations and installations which have been made by customers.	The website will have a dedicated staff “Dashboard” page. This page will allow them to view all upcoming consultations and installation which have been booked by a customer.,
	To be able to add slots for consultations and installations for customer to book.	The website will have a dedicated staff “Slots” page. This page will contain a form allow staff to add available booking slots for customers to book.

Potential Risks:

Risks	How they will be mitigated against
Malicious & Inappropriate Use	For all forms throughout the website input validation will be used. Each form will validate the inputs based on its; • Data type, • Minimum and Maximum value range check, • An array of allowed values (for example days of the week), • Emails will be validated by making sure the email field contains an '@' symbol and a '.' symbol.
User Errors	To help prevent users from making errors the website will: • **Follow design standards and conventions** – This will help the user know what to expect when using the website and will help the website feel more intuitive to use for the first time. • **Clearly Communicates what the user can do** – The website will clearly demonstrate the user what they are and aren't allowed to do. • **Ask users to confirm destructive actions** – The website will have pop-ups asking the user to confirm any destructive actions. The colour of the buttons for these actions will also be different so it stands out against standard actions.
Man in the Middle Attacks (Intercepting Data)	For all passwords field data between transferred from the client-side to the server-side will be encrypted and hashed so prevent it from being vulnerable against Man in the Middle Attacks.
Cross Site Scripting	To help reduce the chance of cross site scripting, as mentioned above, the all input fields will have input validations using the above methods. All input fields will also have input sanitisation to check then modify or remove any potentially harmful data entered by users to prevent against this web-based attack cross site scripting.
SQL Injection	To help reduce the chance of SQL injection, as mentioned above, the all input fields will have input validations using the above methods. All input fields will also have input sanitisation to check then modify or remove any potentially harmful data entered by users to prevent against this web-based attack SQL injection.
Scope Creep	To help prevent against scope creep this proposal document clearly defines the scope of the entire project. This proposal ensures that all stakeholders have a clear understanding of the solutions goals and expected outcomes, as explained above.
Regulatory Compliance	Regulatory and legal requirements and considerations for compliance will be explained below
Delayed Development/Missed Deadlines	To help prevent against delayed development and missed deadlines, this proposal documents gave a priority to both the functional and non-functional requirements. All features will be worked on in priority order as specified above.

Regulatory & Legal Requirements & Considerations:

Requirements	How they will be addressed
Company/Business Details	In the UK, certain information about a company needs to be disclosed on the website and is a legal requirement. To comply with this requirement the website will contain the following information within the footer of the website: • Company name, • Registered number, • Place of registration, • Registered office address, • Contact details, including an email address, As in the brief none of the above information accept the company's name has been given, I will use placeholder text so show compliance and can be switch before the website is hosted.
Privacy Information	To meet the above requirements the website also needs to contain the following: • A privacy notice, • A cookie disclosure, • A disclaimer, As in the brief none of the above documents have been given, I will within the footer of the website show text which can later when the website becomes hosted can be swapped for links to the company's polices.
Cookie Information & Consent	To meet this requirement the website will have a cookie pop up on the website, this will inform them if cookies are used on this site, what they do and why.
Selling to Consumers	As they proposed website is showing products and solutions offered by Rolsa Technologies, the website should also contain a terms and conditions document. As in the brief no terms and conditions document has been given, I will within the footer of the website show text which can later when the website becomes hosted can be swapped for links to the company's terms and conditions document.
Accessibility Requirements	Website must be accessible to all users, including those with disabilities. Complying with the Web Content Accessibility Guidelines (WCAG) helps to ensure inclusivity. The website will follow WCAG by ensure all content is perceivable, operable, understandable and robust, as explained above.
Intellectual Property Rights	The website will respect Intellectual Property Rights by not using any copyrighted materials anywhere on the website without permission. All content on the website will be original, licensed appropriately or solely used for the purpose of the prototype to present to Rolsa Technologies.

Appendix (Research):

Green energy products currently on the market (Heat Pumps):

2. Install a heat pump

Replacing a gas boiler with a heat pump will reduce the carbon footprint of your home by up to 70% – pic credit AIRA

Like a fridge, but in reverse, a **heat pump** is an electrical appliance that transfers heat from the environment outside, and pumps it into the air conditioning, hot water or central heating system inside a building.

Rather than using a fossil fuels to heat water, a heat pump uses heat that has already been generated by solar energy, and uses electricity to pump that heat into our homes and hot water systems.

Heat pumps extract heat from their surroundings. That can be from the air, the ground or a water source outside. This is then used to heat a refrigerant liquid in the heat pump that turns the refrigerant into gas. The gas is then pumped into a compressor, which causes its temperature to rise. The heated gas is then passed over a heat exchange surface from where it can be blown around your home, or transferred into a central heating or hot water system.

A heat pump is an incredibly energy efficient way to heat a building. Installing a heat pump will reduce the carbon footprint of your home by up to 70%.

Owing to the different ways in which they extract heat from their environment, there are various types of heat pump, including **air-source heat pumps (ASHP)**, **ground-source heat pumps (GSHP)**, **water-source heat pumps (GSHP)** and **air-to-air heat pumps**.

As the transition from a gas boiler to a heat pump is expensive, the UK government is currently offering £7500 as a grant towards to the installation costs of a heat pump, through the **Boiler Upgrade Scheme**.

The above image is a screen shot from The Eco Experts website:

https://www.theecoexperts.co.uk/news/8-renewable-energy-ideas-for-your-home#:~:text=panel%20installer%20today.-,2.%20Install%20a%20heat%20pump,-Replacing%20a%20gas

Green energy products currently on the market (Domestic Wind Turbine):

3. Get a domestic wind turbine

*Colourful wind turbines – credit: **Flower Turbines***

The **United Kingdom** is one of the windiest countries on Earth. For this reason, it makes sense to use that free energy for your benefit rather than complain about the weather.

To do this, you'll need your own private **wind turbine**.

We are not talking about the enormous ones that you see on wind farms here, but smaller domestic wind turbines that can be installed on your roof, or freestanding within the grounds around your home.

The electricity is generated by the wind turning the blades of the domestic wind turbine, which drives the internal turbine to create power. An estimated 6kW pole-mounted turbine could save you around £340 a year on electricity bills and earn about £235 per year in **Smart Export Guarantee (SEG)** payments.

The above image is a screen shot from The Eco Experts website:

https://www.theecoexperts.co.uk/news/8-renewable-energy-ideas-for-your-home#:~:text=simple%20form.-,3.%20Get%20a%20domestic%20wind%20turbine,-Colourful%20wind%20turbines

United Nations' Actions for a healthy planet:

https://www.un.org/en/actnow/ten-actions

The above link provides a list of 12 way to lower your carbon footprint and reduce your impact on the environment.

Carbon Interface's Carbon Calculator:

https://www.carboninterface.com/

The above link provides a Carbon Interface Calculator API which I will be able to use in my solution for Rolsa technologies.

W3C WCAG at a Glance:

https://www.w3.org/WAI/standards-guidelines/wcag/glance/

The above link provides an overview of W3C's WCAG which is useful to ensure the website follows the WCAG.

Challenges in Green Technologies – Institute of Sustainability Studies:

https://instituteofsustainabilitystudies.com/insights/lexicon/green-technologies-innovations-opportunities-challenges/#:~:text=Economically%2C%20the%20high%20initial%20investment,and%20implementation%20of%20these%20technologies.

The above link was used to help me gain a better understanding of the challenges the green technologies industry currently faces.

Business Websites – Legal Requirements:

https://www.nibusinessinfo.co.uk/content/business-websites-legal-requirements

The above link was used to help me gain a better understanding of the legal requirements for business's websites.

Energy Cost Calculator:

https://smartmoneytools.co.uk/tools/energy-cost-calculator/

The above link was used to help me gain a better understanding of how a energy cost calculator tool could potentially work for my final solution.